Greenr

Approach for Lloyds PS1: Green Customer Insights Platform

1 Problem and our understanding

Lloyds wants to use customer banking data to find which customers are likely to take up a green product, to estimate the sustainability impact if they do, and to explain both to the customer and to the bank. The brief names four opportunities: EV Finance, Green Home, Solar Finance and Paperless Banking.

2 Idea

Greenr produces one versioned recommendation for every customer and product. Each recommendation holds a calibrated propensity, a Trust score for the quality of the input data, a confidence value that is shown only if it passes validation, reason codes, a recommendation state, and an impact estimate that records its factor source, boundary and assumptions. The customer view, the relationship-manager workspace and the portfolio dashboard all read this same object, and explanations are generated from it, so the wording cannot disagree with the numbers. When the evidence is missing, stale or contradictory, Greenr abstains instead of guessing.

3 Approach

Data. Real bank data is not available, so a seeded generator creates about 4,000 UK customers with roughly 874,000 transactions, plus vehicle, energy, digital-usage and product records. Household energy demand is matched to DESNZ consumption statistics. Adoption is sampled from a monthly probability driven by correlated latent traits, an unobserved preference and noise, so labels are never a threshold on an observed feature. A second, noisier generator with different coefficients is kept aside for stress testing.

Features and models. Each transaction gets exactly one of eight categories, first from merchant rules and then from a TF-IDF and logistic-regression fallback. Customer 360 features are built only from data available before the 30 June 2026 cutoff, and the label is adoption in the following 90 days. Income and postcode are treated as proxy risks: they are excluded from the models and used only for auditing. For each product we compare a hand-written heuristic, a logistic regression and an XGBoost model. Model choice, calibration (Platt or isotonic) and the decision threshold are all fixed on validation data.

Trust, confidence and guardrails. Trust describes the input data: how complete and recent it is, whether sources disagree, and whether the customer is unlike the training population. Confidence is a separate small model that predicts whether a given decision is likely to be correct. It is shown only if higher confidence really does mean fewer errors, and is hidden otherwise. Deterministic rules then place each customer and product in one of six states, such as *recommend*, *insufficient information* or *conflict requires review*.

Impact. Carbon estimates use the pinned DESNZ 2025 greenhouse-gas conversion factors, with the same operational well-to-wheel boundary for the baseline and the scenario; vehicle and battery manufacturing are excluded. Solar is calculated only when the inputs for a PVGIS generation estimate are present, and is otherwise reported as unavailable. No carbon saving is claimed for paperless banking.

Explanations and interface. The language model never sees raw merchant text. It may only arrange pre-approved sentences, and a validator checks that the product, state, numbers, units and disclaimer match the recommendation exactly. If anything fails, a deterministic template is used, so the product works without the model. A Streamlit interface over a FastAPI service serves the customer, the relationship manager and the portfolio team.

4 Testing methodology

All evaluation is on held-out customers. We use a 70/15/15 split by customer, so no customer appears in more than one set, and a third of the validation customers are set aside before any fitting to decide whether confidence can be shown. Test labels are never used to choose a model, a threshold or a release decision.

For each product we report ROC-AUC, PR-AUC, lift in the top 10%, Brier score and calibration error against both the heuristic and the logistic baseline, and we say so when the main model only matches a baseline. Robustness is measured on data from the perturbed generator, with 30% of inputs removed, and across subgroups.

Automated leakage checks stop the pipeline if a customer appears in two splits, a feature is dated after the cutoff, or a label leaks into the features. Categorisation is tested on merchants never seen in training. Confidence is shown only if the most confident 30% of decisions make at most half the errors of the least confident 30%.

Every impact formula has unit tests against hand-calculated values. Explanations are checked programmatically against 200 fixed cases, including prompt injection and requests to guarantee savings or confirm eligibility. Scripted customers cover all six states end to end, and the whole pipeline reruns with fixed seeds from a single `make all`.